ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** Fungi	Hyphae and chitin.
3	**(b)** They are in the same class and phylum	Higher ranks shared.
4	**(b)** animal	Animalia.
5	**(b)** dolphins have lungs, hair and mammary glands	Mammalia vs Pisces.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(d)** Gymnosperms	Cones with naked seeds.
8	**(b)** belong to the same genus but different species	Same genus Canis.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Interbreeding with fertile offspring is the defining feature of a species.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; R states the rule of writing but does not explain why the human name is *Homo sapiens*.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New evidence from molecular studies (1); Discovery of new organisms (1).	1+1
12	Homeothermic: birds and mammals (1); Poikilothermic: fish, amphibia, reptiles (1).	1+1
13	Characteristic of Chordata (1); Used to identify the groups (1).	1+1
14	Archaea live in extreme environments; bacteria in common habitats (1); Archaea have different cell wall and membrane chemistry (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Feathers? yes: pigeon (1); No: has hair? yes: bat (1); No: gills with scales: shark; moist skin: frog; dry scaly skin: snake (1).	1+1+1
16	Moss: gametophyte dominant (1); Fern: sporophyte dominant (1); Both need water for fertilisation (1).	1+1+1
17	Include unrelated organisms (1); Based on a few characters (1); Different evolutionary origins (1).	1+1+1
	OR: Fungi (1); Heterotrophic by absorption (1); Eukaryotic with a cell wall (unicellular like yeast) (1).	1+1+1
18	Fish are the oldest (1); Amphibia, reptiles, birds and mammals evolved later (1); Increasing complexity of heart and brain (1).	1+1+1
	OR: Land animals need to breathe air (1); Shelled egg prevents drying (1); Adaptation to terrestrial life (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) *Mangifera* (1). (ii) *tigris* (1). (iii) No; the species name alone is not enough, they belong to different genera (1). (iv) *Panthera tigris*: genus begins with a capital letter, species with a small letter, and the name is in italics or underlined separately (1); scientific names are the same worldwide and avoid confusion caused by many local names (1).	1+1+1+2
20	Porifera: cellular (1); Cnidaria: radial, tissue (1); Flatworms: bilateral, acoelomate (1); Annelida: segmentation, coelom (1); Chordates: advanced systems (1).	1+1+1+1+1
	OR: Phylum Arthropoda, class Insecta (1); Exoskeleton prevents water loss and protects (1); Wings and jointed legs allow movement (1); Compound eyes and antennae for sensing (1); Importance: pollination, honey, silk, or pests (1).	1+1+1+1+1

Mistakes to avoid:

- Writing scientific names wrongly: *homo Sapiens* or *Homo Sapiens*; the genus has a capital letter, the species a small letter, and both are italicised or underlined separately.
- Treating the kingdom as the smallest group and the species as the largest; the hierarchy runs from kingdom (largest) to species (smallest).
- Putting fungi (mushroom, yeast) in the plant kingdom; they lack chlorophyll and have chitin in their cell wall.
- Mixing up gymnosperms (naked seeds) and angiosperms (seeds enclosed in fruits), or monocots and dicots (one or two cotyledons).
- Saying that all vertebrates are warm-blooded, or that bats (mammals) are birds because they fly.
- Mixing up the phyla: spiders and insects are arthropods, earthworms are annelids, and roundworms are nematodes; also confusing 'Annelida' (segmented) with 'Nematoda' (unsegmented).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** A is true but R is false.	A is true; R is false: they are heterotrophic.
2	**(a)** annelid	Annelida.
3	**(b)** the genus name Panthera	Panthera leo and Panthera tigris.
4	**(d)** Monera	Bacteria.
5	**(b)** it is a bird as it has feathers and lays eggs	Aves.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(b)** mollusc	Mollusca.
8	**(c)** Hominidae	Hominidae.
9	**(c)** A is true but R is false.	Fish, amphibians and reptiles are vertebrates but are cold-blooded, so R is false.
10	**(d)** A is false but R is true.	Fungi lack chlorophyll and are heterotrophs, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Alga and fungus (1); Alga makes food; fungus gives water and protection (1).	1+1
12	Series of choices leading to identification (1); Helps even beginners (1).	1+1
13	They are non-cellular (1); They reproduce only inside host cells (1).	1+1
14	One has two cotyledons (dicot), the other one (monocot) (1); Differences in venation and roots (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Porifera: cellular, asymmetrical, pores (1); Cnidaria: tissue level, radial symmetry (1); Cnidocytes in Cnidaria (1).	1+1+1
16	Honey bee: honey, pollination (1); Silkworm: silk (1); Lac insect: lac; mosquitoes harmful (1).	1+1+1
17	Similar DNA means close relationship (1); Resolves confusions with appearance (1); Led to three domains (1).	1+1+1
	OR: Tiger and leopard more closely related (1); Share the same genus and family (1); Dog is a more distant relative (family Canidae) (1).	1+1+1
18	Decompose dead matter (1); Recycle nutrients (1); Dead matter would accumulate (1).	1+1+1
	OR: Notochord, dorsal nerve cord (1); Pharyngeal gill slits, post-anal tail (1); Present in the embryo, modified later (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Mammalia (mammal) (1). (ii) Aves (bird) (1). (iii) Amphibia (amphibian) (1). (iv) A bat has hair or fur on the body (1) and has mammary glands and gives birth to young ones (any two points) (1).	1+1+1+2
20	Fish: two chambers, single (1); Amphibia: three chambers (1); Reptiles: three with partial division (1); Birds and mammals: four chambers (1); Double circulation (1).	1+1+1+1+1
	OR: Feathers, hollow bones (1); Penguin flippers (1); Bat wings with skin (1); Convergent evolution (1); Different classes (1).	1+1+1+1+1

Mistakes to avoid:

- Writing scientific names wrongly: *homo Sapiens* or *Homo Sapiens*; the genus has a capital letter, the species a small letter, and both are italicised or underlined separately.
- Treating the kingdom as the smallest group and the species as the largest; the hierarchy runs from kingdom (largest) to species (smallest).
- Putting fungi (mushroom, yeast) in the plant kingdom; they lack chlorophyll and have chitin in their cell wall.
- Mixing up gymnosperms (naked seeds) and angiosperms (seeds enclosed in fruits), or monocots and dicots (one or two cotyledons).
- Saying that all vertebrates are warm-blooded, or that bats (mammals) are birds because they fly.

- Mixing up the phyla: spiders and insects are arthropods, earthworms are annelids, and roundworms are nematodes; also confusing 'Annelida' (segmented) with 'Nematoda' (unsegmented).

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** echinoderm	Echinodermata.
3	**(c)** arthropods (insects)	Over a million species.
4	**(d)** classification boundaries are not rigid	Overlap.
5	**(b)** marsupial mammal	Pouch.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(c)** Pisces	Fish.
8	**(a)** angiosperms	Flowering plants.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Plants are autotrophs with cellulose walls; fungi differ in nutrition and cell wall, so they form a separate kingdom.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but one does not explain the other.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They are heterotrophic and multicellular (1); Have no cell wall (1).	1+1
12	Presence of the backbone (1); Invertebrates have many more species (1).	1+1
13	Insects have three pairs of legs, arachnids four pairs (1); Insects have wings and three body parts (1).	1+1
14	Hair and mammary glands (1); Warm-blooded (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Hooks and suckers (1); Thick cuticle (1); High reproductive capacity (1).	1+1+1
16	They interbreed (1); Produce fertile offspring (1); Horse and donkey give sterile mule: different species (1).	1+1+1
17	Platyhelminthes have no body cavity (1); Annelids have a true coelom (1); Cavity gives space for organs (1).	1+1+1
	OR: Bats have hair, not feathers (1); They have mammary glands and give birth (1); Their wings are skin membranes (1).	1+1+1
18	Three pairs of legs, wings (1); Three body parts, antennae (1); Orders: Lepidoptera, Diptera, Hymenoptera (1).	1+1+1
	OR: Linnaeus: plants and animals only (1); Fungi and bacteria put with plants (1); Whittaker: separate kingdoms (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Bryophytes (1). (ii) Pine (a gymnosperm) (1). (iii) Rose (1). (iv) The mushroom is a fungus: it has no chlorophyll and cannot make its own food (1), it feeds on dead matter and its cell wall is made of chitin, not cellulose (1).	1+1+1+2
20	Characters selected by taxonomists (1); Hierarchical ranks (1); New species added (1); Molecular data revise groups (1); Disagreements show the evolving nature (1).	1+1+1+1+1
	OR: High number of endemic species (1); Western Ghats (1); Himalayas (1); Threats (1); National parks and sanctuaries (1).	1+1+1+1+1

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MASTERMIND